

FINDING PICKING POINT USING STRAIGHT LINE DETECTION OF
LABEL ATTACHEMENT MACHINE

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Thesis Finding picking point using straight line detection of label attachment machine

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ABSTRACT

In this study, developed a new technique for detecting the position of labels using a SCARA Robot with the end-effector vacuum pad. The technique first found the image edge with a Canny detector, then selected straight lines and their intersections. It assumed that the label was basically rectangular in shape and roughly aligned so that the key edges were close to horizontal and vertical directions. The slope of the horizontal edge defined the rotational of the label. Otherwise, reducing the image tilting and the distortion by using a perspective transform matrix and thereafter converting the image coordinate into the SCARA robot coordinate. Through the factory environment, the experiment is conducted using various captured image conditions under a lighting source. The performance of the system provides the acceptable accuracy all of the picking coordinates with positioning errors, which x-coordinate are 0.01 – 0.25 mm, y-coordinate are 0.01 – 0.32 mm and rotation angle are 0.05° – 3.94°. In trials, the SCARA robot was able to grip the label precisely in each test.

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Chapter 1

Introduction

In the last decade, the number of competitions for robotic arm design in factory automation has increased extensively. Performance has also increased significantly. However, many types of robots that have different functions. For example, Selective Compliance Assembly Robot Arm (SCARA) robots are lightweight and have small footprints, making them ideal for applications in crowded spaces [1]. Machine vision is the substitution of human visual sense and judgment capabilities with a video camera and computer to perform an inspection task [2]. It is the automatic acquisition and analysis of images to obtain desired data for controlling or evaluating a specific part or activity [3]. Robotic arms using machine vision techniques have been used wisely in industry for the location and orientation of the object, then commanding the robot device to the expected coordinates to grip the workpiece.

1.1 Problem Description

There are two steps in Label attachment, the label barcode must be transmitted the data to the server and the label attached to a printed circuit board (PCB), which has a minimum label size of 6×6 mm. There are some errors in manual processes, - missing scanned barcode data to be sent to the server and wrong alignment of label when attaching it, because of low visibility of the labeling box as shown in figure 1.1. Moreover, the production investment cost in the manpower is raised for the long-term future. Robotic systems can automate processes to reduce human lack of attention during work, reduce cycle and lead time, and expenses. Automated machines can achieve better performance and high precision in production lines, which led to a desire to develop the label attachment machine, see figure 1.2. Here, in the study

focused on finding the positions of labels, using simple fast image processing, to design software to integrated with a SCARA robot.

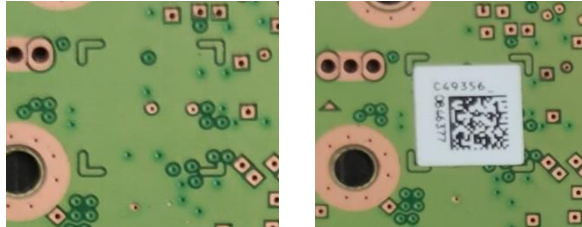


Figure 1.1: Labeling box

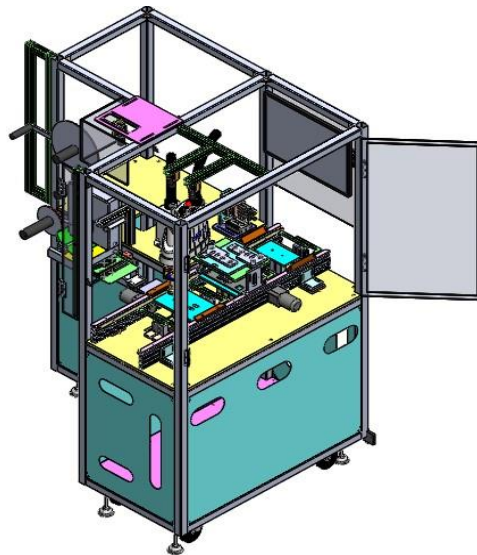


Figure 1.2: Label attachment machine drawing

1.2 Research Objectives

1. To study the principle of SCARA robot, machine vision system and PLC controller.
2. To study working of integrated machine vision and SCARA robot.
3. To study the impact of Machine vision software, finding the coordinate of gripping labels.
4. To apply the Label attachment machine into production lines.

1.3 Scope of the study

1. Design PLC program to control and motion of SCARA robot and intermediary communicating between robot and vision method.
2. Design machine vision method which is able to detect labels and find axis coordinates precisely.
3. Design operation methods of Label attachment machine.

1.4 Expected Contribution

1. Label attachment machine able to apply to work in production lines.
2. SCARA robot is able to pick labels in precise coordinates
3. To improve knowledge of working integrated machine vision systems and robotics.
4. To develop and understand communicating between Machine vision, Epson Robot, Visual basic application and PLC controller.

Chapter 2

Theory and Literature Review

2.1 SCARA Industrial Robotic

The usage of robotic arms increased enormously in the industrial automation market and various manipulators such as cartesian, articulated arm, polar, cylindrical, etc. In general, industrial robots are employed to carry out repetitive jobs and/or those that require precision and speeds difficult to achieve by human beings. On the other hand, industrial robots can perform tasks during many hours per day without getting tired or losing precision or effectiveness, because they are currently highly developed and robust devices that practically do not fail [4]. Thus, to know, study, improve, reprogram, and adequate these systems to different scenarios becomes necessary, so users can profit as much as possible from them.

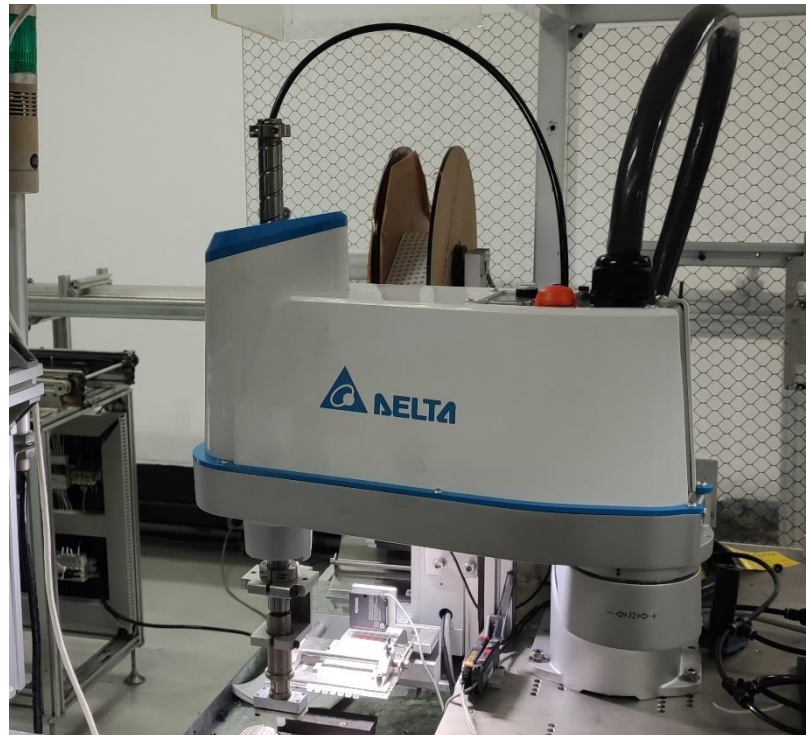


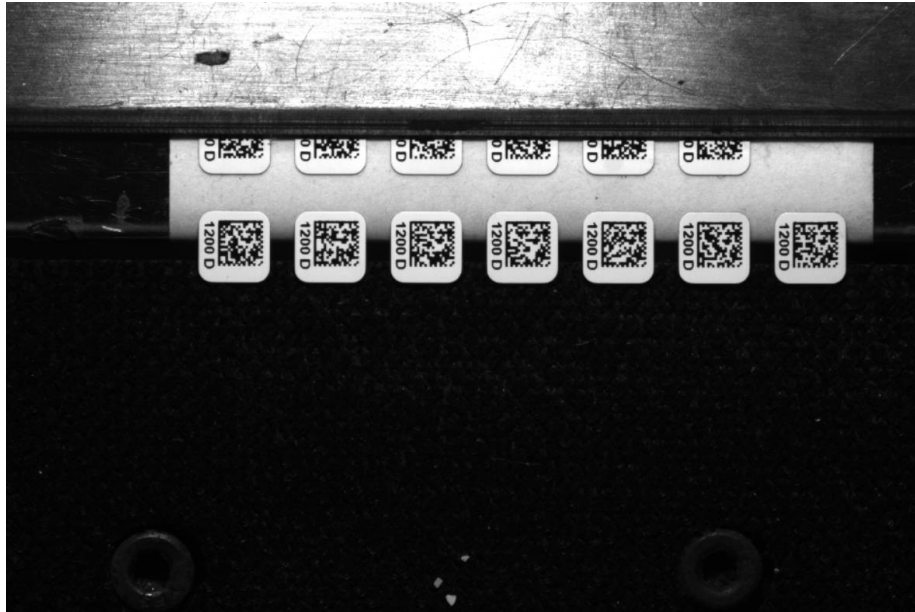
Figure 2.1: SCARA robot

SCARA industrial robot, see in figure (2.1), with a high degree of flexibility and versatility has been in the electronics, automotive, and other industries access to a wide range of applications in process automation. It is a highly non-linear, strong coupling, time-varying system [5]. SCARA normally has 4- DOF in which one is linear motion and three rotational movements. It is very common to use for pick and place applications including the assembly task where is capable of very fast cycle times.

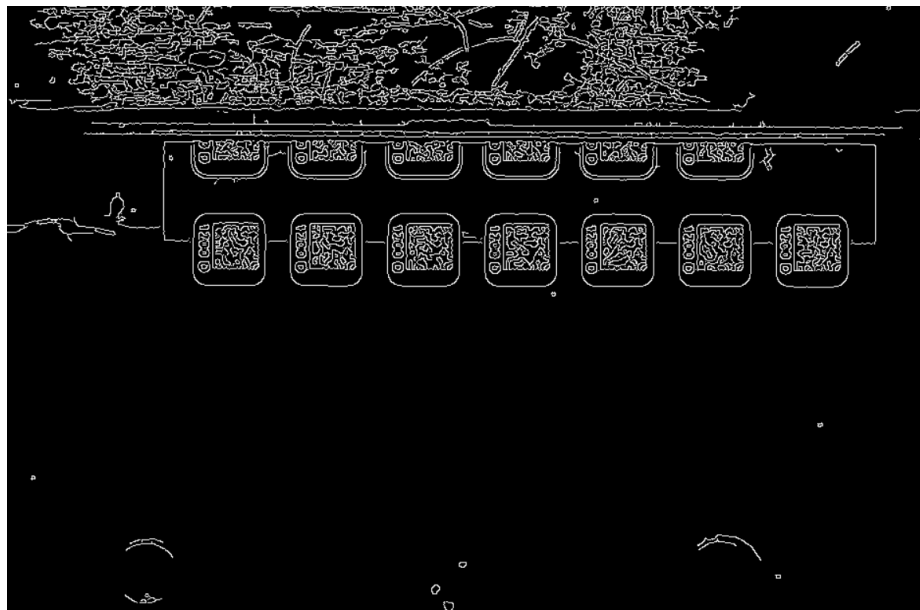
2.2 Edge Detection

Edge Detection is an approach frequently used in segmenting pictures based on the change in sharp intensity of the picture. The definition of edge in this context is a set of connected-pixel located in two areas [6]. Edge detection is the important step in object extraction. Therefore, it becomes important to know about various edge detection techniques. Amer and Abushaala [7] studied some edge detection based which the result show that Gradient based edge detection algorithms like Robert's algorithm: have a major drawback of being very sensitive to noise and produces weak responses for genuine edges, Sobel's algorithm: less sensitive to noise but computationally it is slower, Prewitt's algorithm: the response of the prewitt kernel is too small and further processing has to be done. Different edge detection methods can be implemented as per the need of segmentation of image. Canny edge detection algorithm is less sensitive to noise. However, Canny operator performed better than Sobel, Prewitt, Roberts and LOG.

Canny edge detection is a popular edge detection algorithm. It was developed by John F. Canny in 1986 which is a multi-step algorithm [8]. This algorithm uses an optimal edge detector based on a set of criteria which include finding the most edges by minimizing the error rate, marking edges as closely as possible to the actual edges to maximize localization, and marking edges only once when a single edge exists for minimal response [9].



(a) Original image



(b) Canny edge image

Figure 2.2: (a) Original image, (b) Canny edge image

2.3 Hough Line Transform

One of the main approaches to describing objects is by using mathematical functions that describe lines effectively called Hough Transform, named by Paul Hough, who invented and patented this method in 1962. The Hough Transform is the

transformation between the Cartesian coordinate and the space of another variable which is a straight line (or the definition of a boundary in the form of a higher-order curve) can be defined [10]. Consider the equation of a normal line in the Cartesian coordinate system.

$$y - mx - c = 0 \quad (2,1)$$

The Hough line transform apply $(\theta - \rho)$ as representation of a straight line in (x, y) image space

$$\rho = x.\cos\theta + y.\sin\theta \quad (2.2)$$

Here, θ is the angle of perpendicular make with the x-axis and ρ is the distance between line and origin. We can consider the corresponding $(\theta - \rho)$ value and calculate by equation (2.2) into the polar coordinate called Hough space, also known as accumulator point array $H(\rho, \theta)$. For every pixel, votes $H(\rho, \theta)$ are setting zero at the beginning and increase the votes by 1 in the cell representing (ρ, θ) .

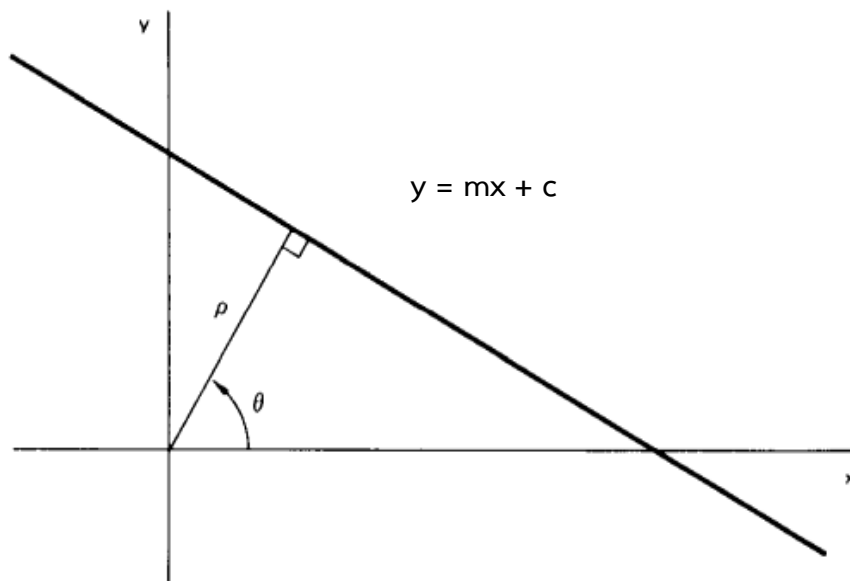


Figure 2.3: parameters for a line [11]

Basic Hough transform algorithm

1. Initialize $H(\rho, \theta) = 0$
2. For each edge pixel of image (x, y)

For $\theta = [\theta_{min} \text{ to } \theta_{max}]$

$$\rho = x.\cos\theta + y.\sin\theta$$
 Accumulator point: $H(\rho, \theta) += 1$
3. Find the value of (ρ, θ) where $H(\rho, \theta)$ is maximum
4. The detected line in the image is given by $\rho = x.\cos\theta + y.\sin\theta$

2.4 OpenCv Software

Specially designed software, OpenCv [12], are being used for image processing. OpenCV (Open-source Computer Vision) is an open-source library which contains more than 500 algorithms optimized for pictures and video analysis. Since it was introduced 1999, most of it had been adopted as main development means for the society, researchers and developers in image processing and computer vision. Emgu CV (<https://www.emgu.com>) is a cross platform .Net wrapper to the OpenCV image processing library. Allowing OpenCV functions to be called from .NET compatible languages. The wrapper can be compiled by Visual Studio and Unity, it can run on Windows, Linux, Mac OS, iOS and Android. EMGUCV [13] is written entirely in C#. It can be run on any platform .net supports, including iOS, Android, Mac OS X, Linux and windows. A lot of efforts have been spent to have a pure C# implementation since the headers have to be ported, compared with managed C++ implementation where header files can simply be included. But it is well worth it if you see EMGUCV running on Fedora 10! Pluses it always gives you the comfort knowing that your code is cross-platform.

2.5 Related paper

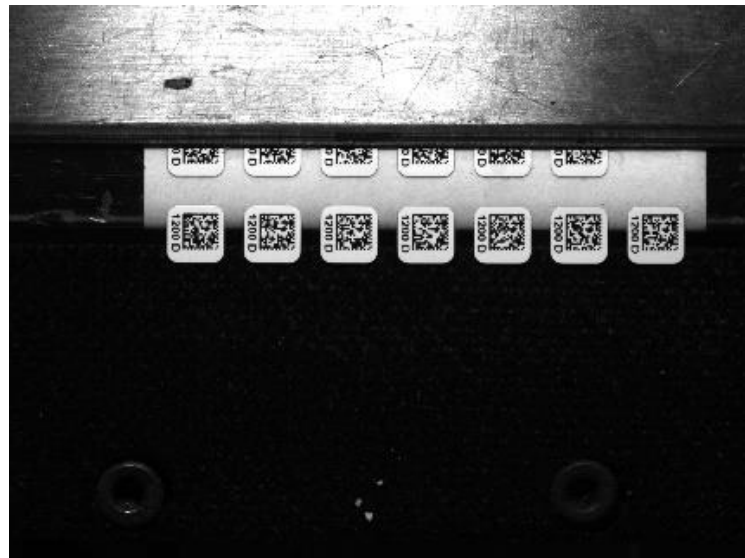
Many studies have discussed robotic pick and place systems. Nerakaea, Uangpairajb, and Chamniprasart designed a prototype of a flexible automatic assembly system combined with a machine vision system. Using LabVIEW NI vision software and NI vision builder, the captured image was used to measure the exact target position for placing the assembly part. They calibrated their system by using image, which has a size standard of circle Forty-eight circles (6x8 rows) with 10 mm diameter in its application [14]. Huang, Lin, and Chen used a scheme to measure the relative distance between the object and robot arm using edge detection in captured images [15]. Its process involves the environment indoor and hall, and the color differentiating of input visualization from the camera to locate the candidate block of fitting color requirements and thereafter define the sector. Kim et al [16] described vision-based bin picking system using multiple local features for an industrial robot. The setting of the working environment for bin-picking, the detection of object location, pose estimation, distance measurement and surface orientation angle detection steps were included in their system. Pyo et al [17] described a vision-based angle estimation algorithm for bin picking. They described a rotation angle estimation method, which was not influenced by the pose of the object and used an affine transformation to preprocess the image for invariance to object posture. Hu et al. presented a vision-based SCARA system for spanner fetching and placing, which can be applied in traditional factories [18]. They used a SCARA robot with a 500 mm working range, and four degrees of freedom. It was designed to be low-cost. Farag, Abd Ghafar and Alsibai [19] aimed to increase the impact of computer vision on robotic localization and grasping applications. They implemented comparison of edge detect methods, object detection and localization to enable robotic grasping and positioning using SCARA.

However, the entire working environment in the factory has different lighting conditions to achieve better precision of vision processing.

Chapter 3

Methodology

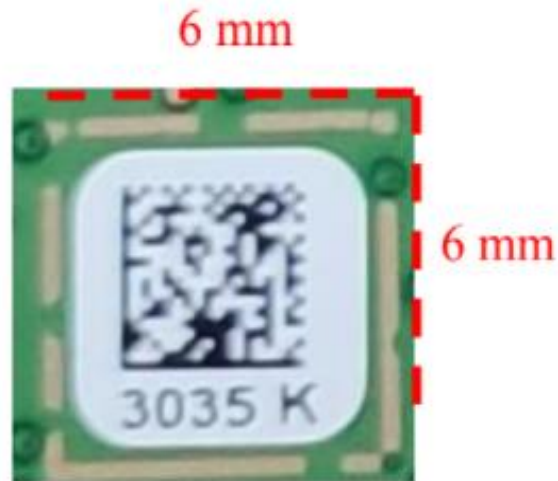
In this chapter, the methodology followed in the work is explained deeply. The method used is explained in detail about detecting the labels, localization and orientation of picking coordinates, image and robot coordinate transformation and experiment results. The label size is 6 mm diameter, see the figure (3.1).



(a) Label at feeder (picking position)



(b) Label attached to PCB



(c) Size of label on attachment

Figure 3.1: (a) Label at Feeder (Pick Position), (b) Label PCB attachment (Place Position), (c) Size of label on attachment

3.1. Hardware device systems

The basic hardware for label localization [10] contained four components, a robot, industrial camera, lighting source, and industrial personal computer. The robot arm was a Delta SCARA robotics model of DRS60L6SS1BN002: the single camera was Basler model 14GM-2500AC, with 50 mm lens, resolution size of 5M pixels, and 14 fps, seen in the acrylic box above and at the right-hand of the robot. The SCARA robot performs the picking and attaching object operation. An added light source maintained stable lighting for a captured images and adapting to the factory environment. The height from camera to the target object was approximately 750 mm, allowing space for the robot to pick the object comfortably. A 5.5 mm vacuum pad was attached to the end effector of the robot. Images were captured and processed on the industrial computer, Nuvo-6032 Window 7 Intel Skylake, with 6th Gen core. The system is shown in Figure (3.2).

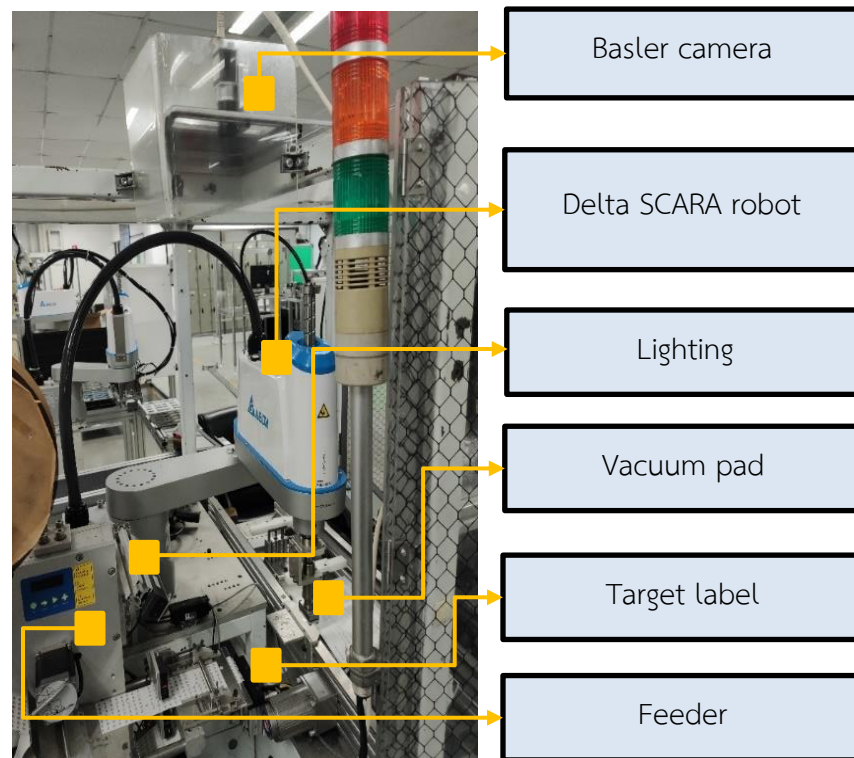


Figure 3.2: Hardware device systems

3.2. Algorithm for finding label position

The positioning algorithm was programmed by EMGUCV, a cross-platform .Net wrapper to the OpenCV, an image processing library, which captured the image from the Programmable Logic Controller (PLC) via TCP/IP. The PLC was an intermediary for communicating and controlling output devices that was programmed by a ladder diagram, and stored various data, including labels, status values, etc. The label coordinate is defined through the data flow as shown in figure (3.3).

An image was captured, transformed the image to the grayscale and conduct the edge detection image. Then the seven labels in each image was detected. If they were not detected, a status message was sent to the PLC “no label found”. Then we search for the next label. The label detection algorithm, finding the left edge as a vertical line and the bottom edge as a horizontal line using Hough transform. Intersecting both vertical and horizontal lines make a “gripping point” to transform the image coordinates to robot coordinates. The label angle was determined from the horizontal

line. Then the label position was sent to the robot, and responding to the PLC with a “label found” message.

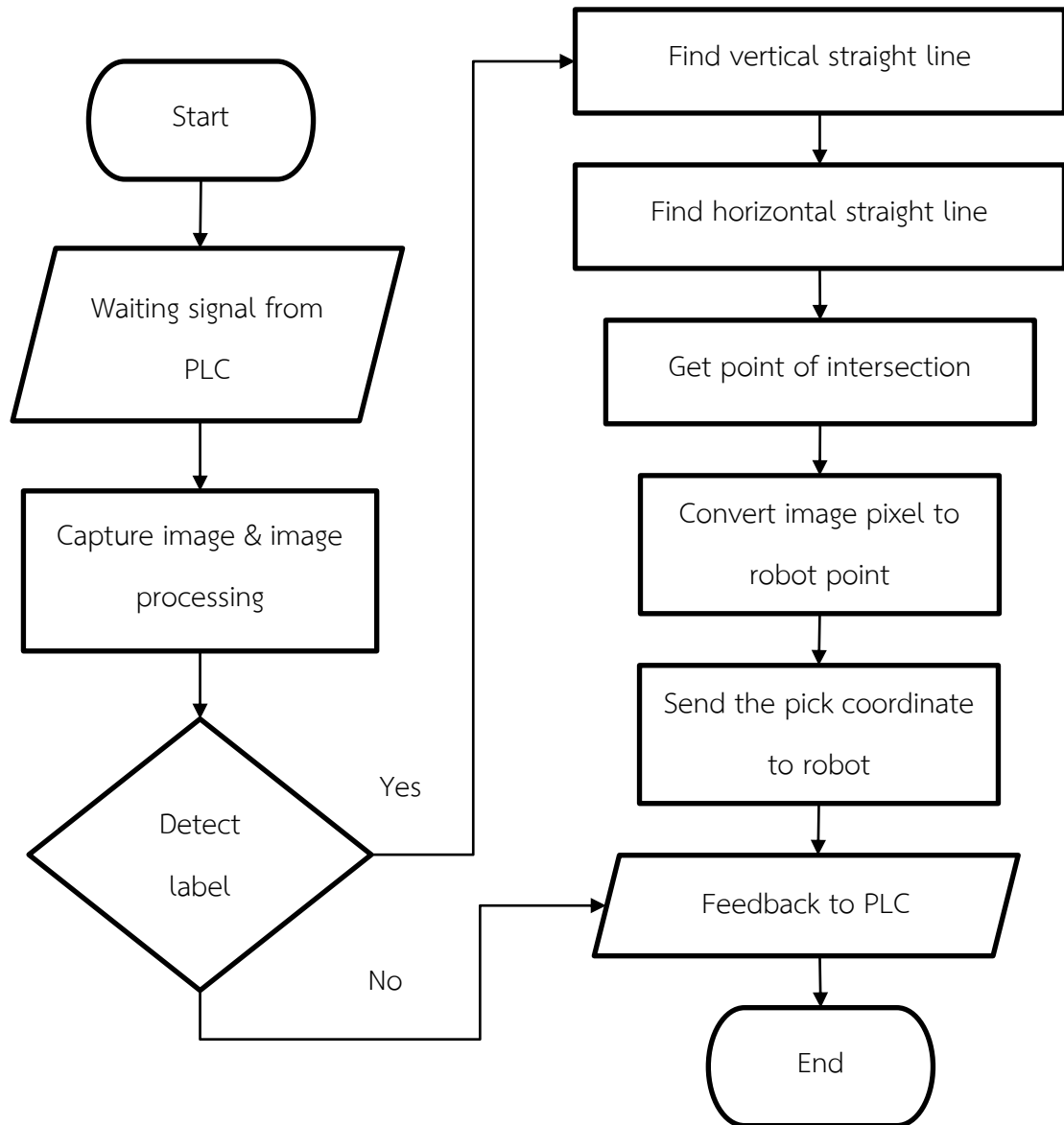


Figure 3.3: Algorithm for finding the label position

Chapter 4

Experiments And Results

In this section, the method used is explained in detail about detecting the labels, localization and orientation of picking coordinates, image and robot coordinate transformation and experiment results.

4.1. Experiment methods

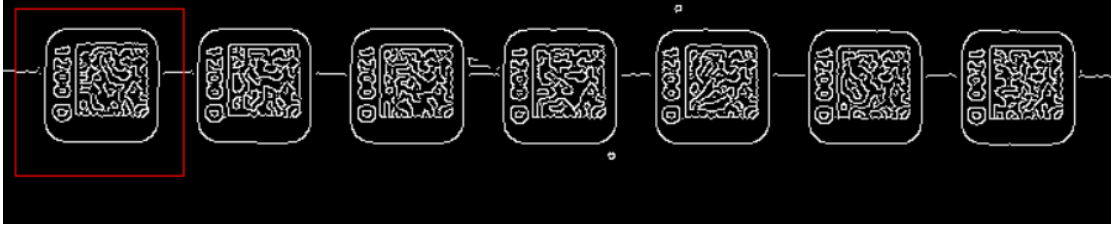
4.1.1. Detecting the Labels

From the shown image figure (3.3), the captured 2592×1944 pixels image was converted to grayscale, see figure (4.1 - a), where the label was on the bar of the feeder. The approach used an OpenCV GaussianBlur function with kernel size of 5×5 , a canny edge detector and detected intersections between the outer edges of the label to find the picking point.

There are three steps to find picking points in the grayscale image. First, a canny edge detector with 5×5 kernel was applied to find out approximate shapes of objects. Then an ROI for the first label was set as input to eliminate some extraneous noise and speed up computation - see the red rectangle in figure (4.1 - b). Known differences to the subsequent labels were used to set ROIs for them. Inside these ROIs, the OpenCV FindContour function was used to find potential contour: the moments of these contours identified a rectangle of suitable size and its center point was set the reference to point of the labels. Noise inside the label was filled up as black to remove some undesired scattered white. Figure (4.1 - c) shows the image after removing noise. Finally, straight lines were detected by a Hough transform, where finding vertical and horizontal lines from the edges of images. The intersection points between vertical and horizontal lines provided the picking point, while the angle obtained from the horizontal line, see figure (4.1 - d).



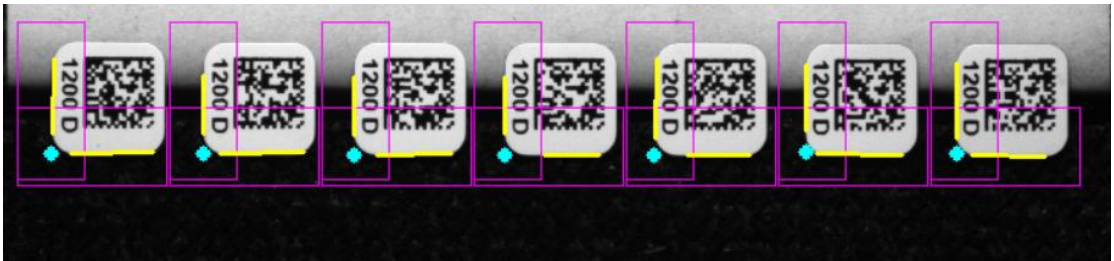
(a) Grayscale image



(b) Edge image



(c) Detected label & filled image



(d) Vertical and horizontal line detection and intersection point

Figure 4.1: a) Grayscale Image, b) Edge image, c) Detected label & Filled image, d) Vertical and horizontal line detection and Intersection point, all images shown are only a part of the full image.

4.1.2. Localization and orientation of picking coordinates

Both labels and the vacuum suction head attachment are rectangular in shape, where line L_1 is defined by points (x_1, y_1) and (x_2, y_2) , and line L_2 is defined by (x_3, y_3) and (x_4, y_4) , the intersection (x, y) is given by equation (4.1)

and (4.2). The orientation of the label, given by the horizontal line L_2 , is computed by equation (4.2).

$$x = \frac{\begin{vmatrix} |x_1 & y_1| & x_1 - x_2 \\ |x_2 & y_2| & x_1 - x_2 \\ |x_3 & y_3| & x_3 - x_4 \\ |x_4 & y_4| & x_3 - x_4 \end{vmatrix}}{\begin{vmatrix} |x_1 - x_2 & y_1 - y_2| \\ |x_3 - x_4 & y_3 - y_4| \end{vmatrix}} \quad (4.1)$$

$$y = \frac{\begin{vmatrix} |x_1 & y_1| & y_1 - y_2 \\ |x_2 & y_2| & y_1 - y_2 \\ |x_3 & y_3| & y_3 - y_4 \\ |x_4 & y_4| & y_3 - y_4 \end{vmatrix}}{\begin{vmatrix} |x_1 - x_2 & y_1 - y_2| \\ |x_3 - x_4 & y_3 - y_4| \end{vmatrix}} \quad (4.2)$$

$$Angle = \arctan \left(\frac{y_3 - y_4}{x_3 - x_4} \right) \quad (4.3)$$

4.1.3. Image and robot coordinate transformation

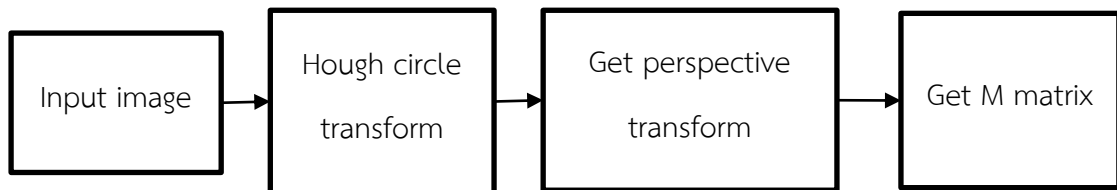
A perspective Transformation [21] is to reduce the image tilting and distortion caused by the 3D world position obtained by the camera to 2D plane coordinates [22]. The transformation matrix is equation (4.4).

$$\begin{bmatrix} x_p \\ y_p \\ 1 \end{bmatrix} = M \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (4.4)$$

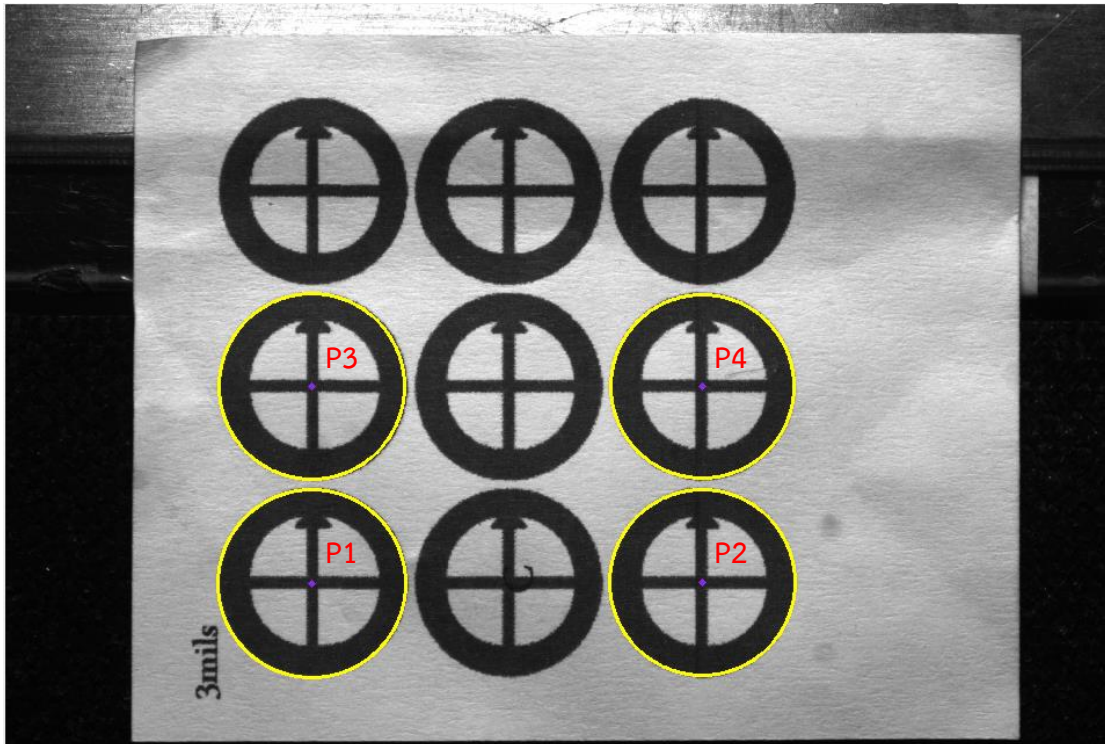
Where the transform matrix M , see equation (4.5), was derived from the four center points of circles in the image calibration.

$$M = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix} \quad (4.5)$$

To calculate the conversion matrix, first a Canny edge detector was used to find pattern edges and Hough transform, found the circle centers – see the four points, P1, P2, P3, and P4, in the calibration image – seen figure (4.2 - b). Using the OpenCv `getPerspectiveTransform` function to compute the M matrix using the steps in figure (4.2 - a).



(a) Calibration and coordinate conversion method



(b) Calibration and Conversion Coordinate Image

Figure 4.2: (a) Calibration and coordinate conversion method, and (b) Calibration and Conversion Coordinate Image

After obtaining the M matrix, the target intersection (x_p, y_p) , for the robot was calculated by computing the determinant the equation (3), thus the result is an equation (4.6) and (4.7).

$$x_p = \frac{m_{11}x + m_{12}y + m_{13}}{m_{31}x + m_{32}y + m_{33}} \quad (4.6)$$

$$y_p = \frac{m_{21}x + m_{22}y + m_{23}}{m_{31}x + m_{32}y + m_{33}} \quad (4.7)$$

Table (4.1) show the result of a calibration and conversion matrix for converting pixel position to the world coordinates (mm) using equation (4.6) and (4.7) where the corresponding four-points see the figure (4.2 – b).

Table 4.1: Four points corresponding and resultant M matrix

Coordinate Location	Image (pixel)		World (mm)	
	X (image)	Y (image)	X (robot)	Y (robot)
P1	1087	1152	132.86	-283.56
P2	1422	1150	132.67	-255.98
P3	1085	981	118.72	-283.42
P4	1421	981	118.82	-255.98
M	$\begin{bmatrix} 0.00403 & 0.09423 & 30.63460 \\ 0.08097 & -0.01092 & -381.8564 \\ 0.00003 & 0.00004 & 1 \end{bmatrix}$			

4.2. Experiment Results

In the experiment, we captured 12 rows of different label images with various characters as shown in figure (4.3) to test performance. Each label image was tested 30 times. We evaluated the results by measuring the actual positioning and rotation

angle of robot coordinates with the computed positions of image coordinates in each row of labels using equation (4.8).

$$Errorr = |P_{robot} - P_{computed}| \quad (4.8)$$

where P_{robot} is an actual robot point and $P_{computed}$ is a computation was measured from image coordinate, the tests were run with and without the additional light, using the ambient laboratory fluorescents alone.



Figure 4.3: Partial experiment results with the rotation angle of label

Table (4.2) show the average, minimum and maximum positioning error for label position 1 to 7. The overall of coordinate error sorted in order (x, y, angle) where the average error was (0.11 mm, 0.07 mm, 0.90 °), the minimum error was (0.01 mm, 0.01 mm, 0.05 °) and the maximum error was (0.25 mm, 0.32 mm, 3.94 °).

For this robot, the vacuum head was 5.5×5.5 mm in size, whereas the targets where 6×6 mm, so an acceptable tolerance was 0.5 mm. Hence, our results show that the robot should be able to grip the label every time.

Table (4.3) shows the result of each position of labels. With the light source, all label position was detected with a 100%. The detected rate without the light source, label positions 1 to 4 was 83% and the following label positions 5, 6 and 7 were 75%,

41% and 25%. The computed time of seven position of labels including the edge detection, finding straight line and coordinate transforming was ~180 ms.

Table 4.2: The experiment result of positioning error

Position	Error	x-error (mm)	y-error (mm)	Angle-error (degree)
1	AVERAGE	0.11	0.04	0.92
	MIN	0.03	0.01	0.09
	MAX	0.18	0.08	3.88
2	AVERAGE	0.07	0.07	1.41
	MIN	0.01	0.02	0.09
	MAX	0.19	0.17	3.94
3	AVERAGE	0.09	0.10	0.83
	MIN	0.01	0.03	0.06
	MAX	0.22	0.27	1.43
4	AVERAGE	0.09	0.04	0.88
	MIN	0.01	0.02	0.06
	MAX	0.16	0.12	1.22
5	AVERAGE	0.09	0.07	0.76
	MIN	0.01	0.01	0.06
	MAX	0.25	0.29	2.60
6	AVERAGE	0.13	0.12	0.75
	MIN	0.02	0.02	0.05
	MAX	0.23	0.29	2.15
7	AVERAGE	0.16	0.09	0.78
	MIN	0.09	0.02	0.06
	MAX	0.23	0.92	2.39
AVERAGE		0.11	0.07	0.90
MIN		0.01	0.01	0.05
MAX		0.25	0.32	3.94

Table 4.3: Detected label with and without the light source

Label Position \ Detected rate	Detected rate		Computed time (ms)
	with light (%)	without light (%)	
1	100	83.33	~180
2	100	83.33	
3	100	83.33	
4	100	83.33	
5	100	75.00	
6	100	41.67	
7	100	25.00	

Chapter 5

Conclusion

In this thesis, the system provided sufficiently accurate picking coordinates for the SCARA robot to grip the label effectively. Here, we developed the label attachment machine which focused on the picking point. Our method relied on straight line detection to locate a picking point and to position the 6 mm diameter labels in a machine system integrated into a SCARA robot. Moreover, a perspective transform allowed labels to be rotated relative to the camera axis. In addition, the defined four points of the robot are important for transforming coordinates in order to achieve accuracy. Through the factory environment, the proper light source is also important for the proposed system.

For future study, a conveyor tracking will be studied to upgrade the system at the robot placing side, after gripping the label, the robot could be attached the label to the PCB while it is moving on the conveyor with the various model changing automatically.

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APPENDIX A

Experiments testing data collection

Table A-1-1: Robot of X-Coordinate (Actual point)

Label \ Point	Image X-Coordinate (mm)						
	X1	X2	X3	X4	X5	X6	X7
1	115.709	115.698	115.728	115.725	115.725	115.864	115.823
2	115.539	115.698	115.698	115.725	115.615	115.700	115.825
3	115.755	115.669	115.798	115.725	115.715	115.702	115.802
4	115.710	115.671	115.660	115.723	115.744	115.883	115.821
5	115.491	115.531	115.531	115.720	115.634	115.625	115.735
6	115.493	115.533	115.529	115.635	115.816	115.620	115.735
7	115.750	115.705	115.813	115.815	115.817	115.826	115.835
8	115.447	115.552	115.495	115.641	115.717	115.763	115.734
9	115.717	115.521	115.678	115.639	115.746	115.736	115.716
10	115.771	115.783	115.972	115.641	115.717	115.763	115.934
11	115.644	115.291	115.281	115.559	115.671	115.739	115.352
12	113.329	115.618	115.789	115.935	116.101	116.258	116.766

Table A-1-2: Image of X-Coordinate (Computed point)

Label \ Point	Image X-Coordinate (mm)						
	X1	X2	X3	X4	X5	X6	X7
1	115.622	115.661	115.793	115.808	115.848	115.795	115.910
2	115.631	115.644	115.684	115.840	115.868	115.903	116.008
3	115.602	115.662	115.701	115.837	115.847	115.904	116.033
4	115.603	115.633	115.678	115.835	115.866	115.903	116.008
5	115.519	115.560	115.684	115.629	115.781	115.790	115.926
6	115.422	115.543	115.584	115.629	115.764	115.790	115.829
7	115.620	115.627	115.767	115.824	115.832	115.874	116.008
8	115.519	115.578	115.580	115.724	115.764	115.886	115.926
9	115.547	115.334	115.583	115.741	115.847	115.886	115.817
10	115.588	115.704	115.750	115.794	115.821	115.952	116.090
11	115.486	115.171	115.147	115.395	115.614	115.513	115.211
12	113.441	115.815	115.933	116.038	116.08	116.216	116.573

Table A-2-1: Robot of Y-Coordinate (Actual point)

Label \ Point	Robot Y-Coordinate (mm)						
	Y1	Y2	Y3	Y4	Y5	Y6	Y7
1	-288.312	-280.237	-272.097	-264.104	-255.980	-247.849	-239.743
2	-288.458	-280.384	-272.199	-264.104	-256.113	-248.050	-239.918
3	-288.492	-280.186	-272.231	-264.225	-256.163	-248.048	-239.567
4	-288.458	-280.385	-272.192	-264.258	-256.126	-248.105	-239.988
5	-288.455	-280.484	-272.292	-264.155	-256.122	-248.055	-239.920
6	-288.585	-280.485	-272.295	-264.329	-256.124	-248.105	-240.110
7	-288.584	-280.383	-272.422	-264.328	-256.124	-248.051	-239.875
8	-288.587	-280.576	-272.406	-264.329	-256.124	-248.116	-239.775
9	-288.909	-280.300	-272.406	-264.239	-256.289	-248.115	-239.948
10	-288.587	-280.577	-272.406	-264.329	-256.124	-248.116	-239.975
11	-288.725	-280.826	-272.965	-264.530	-256.478	-248.515	-240.372
12	-290.192	-280.585	-272.276	-264.166	-256.289	-248.297	-239.535

Table A-2-2: Image of Y-Coordinate (Computed point)

Label \ Point	Image Y-Coordinate (mm)						
	Y1	Y2	Y3	Y4	Y5	Y6	Y7
1	-288.312	-280.237	-272.097	-264.104	-255.980	-247.849	-239.743
2	-288.458	-280.384	-272.199	-264.104	-256.113	-248.050	-239.918
3	-288.492	-280.186	-272.231	-264.225	-256.163	-248.048	-239.567
4	-288.458	-280.385	-272.192	-264.258	-256.126	-248.105	-239.988
5	-288.455	-280.484	-272.292	-264.155	-256.122	-248.055	-239.920
6	-288.585	-280.485	-272.295	-264.329	-256.124	-248.105	-240.110
7	-288.584	-280.383	-272.422	-264.328	-256.124	-248.051	-239.875
8	-288.587	-280.576	-272.406	-264.329	-256.124	-248.116	-239.775
9	-288.909	-280.300	-272.406	-264.239	-256.289	-248.115	-239.948
10	-288.587	-280.577	-272.406	-264.329	-256.124	-248.116	-239.975
11	-288.725	-280.826	-272.965	-264.530	-256.478	-248.515	-240.372
12	-290.192	-280.585	-272.276	-264.166	-256.289	-248.297	-239.535

Table A-3-1: Robot of Angle-Coordinate (Actual point)

Label \ Point	Robot Angle-Coordinate (degree)						
	Y1	Y2	Y3	Y4	Y5	Y6	Y7
1	0.086	0.140	0.14	0.056	0.056	0.056	0.056
2	0.086	0.140	0.14	0.056	0.056	0.056	0.056
3	0.086	0.094	0.094	0.056	0.056	0.056	0.056
4	0.086	0.094	0.094	0.056	0.056	0.056	0.056
5	0.086	0.094	0.094	0.056	0.056	0.056	0.056
6	0.086	0.094	0.094	0.056	0.056	0.056	0.056
7	0.086	0.094	0.094	0.056	0.056	0.056	0.056
8	0.086	0.094	0.094	0.056	0.056	0.056	0.056
9	2.990	-0.730	0.094	0.056	0.056	0.056	-0.857
10	0.086	0.086	0.094	0.056	0.056	-1.056	0.056
11	-2.632	-1.032	-6.211	-3.268	-2.331	-4.335	-3.490
12	-30.449	-5.095	0.056	-0.666	-0.340	0.086	13.467

Table A-3-2: Image of Angle-Coordinate (Computed point)

Label \ Point	Image Angle-Coordinate (degree)						
	Y1	Y2	Y3	Y4	Y5	Y6	Y7
1	0.000	-2.290	1.169	1.080	0.000	1.060	1.273
2	1.218	0.000	0.000	1.280	1.101	1.041	0.000
3	0.000	1.060	1.041	1.193	0.000	1.060	1.169
4	0.000	0.000	-1.218	0.000	1.080	0.000	-1.145
5	0.000	0.000	0.000	-1.060	1.060	-1.123	0.000
6	-1.060	-1.160	-1.123	-1.060	0.000	-1.123	-1.060
7	1.060	-1.145	0.000	1.060	-1.145	-1.080	0.000
8	0.000	1.060	-1.332	0.000	0.000	0.000	0.000
9	1.301	-4.666	-1.123	1.041	0.000	0.000	-2.161
10	-1.101	-1.363	-1.145	-1.060	-2.544	-1.101	0.000
11	-3.179	-4.316	-7.386	-4.484	-3.652	-6.483	-4.184
12	-34.33	-6.203	0	-1.145	-1.06	0	11.07

APPENDIX B

Screenshots of the testing application

